Decomposition:

$$\langle \tau_{neat}(c) \rangle = f_W \tau_W^{cc}(c) + (1-f_W)\tau_A^{cc}(c)$$

$$\frac{1}{\tau_X(c)} = \frac{1}{\tau_X^{diff}(c)} + \frac{1}{\tau_X^{jump}(c)}, \quad X \in \{A, W\}$$

f_W bulk (pure water); $f_W = 1 \Rightarrow \langle \tau_{neat} \rangle = \tau_W(c) = \tau_W^{bb}$

$\delta\tau :=$ (Avg. neat @ conc. $= c$) − (Avg. neat in bulk)

$\delta\tau = \langle \tau_{neat}(c) \rangle - \tau_W^b(0) = f_W \tau_W^{cc} + (1-f_W)\tau_A^{cc} - \tau_W^{bb}$

↓ Decomposition

$$\delta\tau(c) = \delta\tau_p(c) + \delta\tau_f^W(c) + \delta\tau_f^A(c) + \delta\tau_j^W(c) + \delta\tau_j^A(c)$$

$\delta\tau_p(c)$: This talks about the overall change in the state of H-bonds; for $f_W = 1$ to $f_W < 1$. Slowness due to OH being bound to anions.

$\delta\tau_f^W(c)$: Change in overall RotO time of O−O frame; keeping jump time fixed.

$\delta\tau_f^A(c)$: Change in the overall RotO time of O−A frame; (A-bound) keeping jump fixed.

$\delta\tau_j^W(c)$: Change in overall RotO time due to O→O/A jump; keeping frame fixed.

$\delta\tau_j^A(c)$: Change in overall RotO time caused by change in the jump times from A→O or A→A; time b/w fixed.

(Often times talks about individual changes i.e. jump change for 1 ion in bulk w/s @ conc. $= c$)

$$\tau_X^{ij} : (i,j) \in \{c, b\}^2, \quad X \in \{A, W\}$$

→ computed using EJM

$$\rightarrow \delta\tau_\rho = \frac{1-\rho_W}{2}\left[\left(\tau_A^{bb} - \tau_W^{bb}\right) + \left(\tau_A^{cc} - \tau_W^{cc}\right)\right]$$

diff. btw minimal RED in bulk of waters and RED in bulk

diff. btw amino acid @ conc. = c and waters RED @ conc. c

Aside:

$$\frac{1}{\tau_X^{ij}} = \frac{1}{\tau_X^{jump}(ij)} + \frac{1}{\tau_X^{frame}(j)} \quad ; X \in \{A, W\}$$

$i, j \in \{b, c\}$

$\forall i \neq j : \tau_X^{ij}$ is an imaginary constant.

$$\rightarrow \delta\tau_f^W(c) = \frac{1+\rho_W}{4}\left[\left(\tau_W^{cc} - \tau_W^{bb}\right) + \left(\tau_W^{cc} - \tau_W^{cb}\right)\right]$$

change in 0-0 frame @ c from bulk when jumping @ bulk conditions

change in 0-0 frame @ c from bulk when jumping @ concentrated conditions

$$\rightarrow \delta\tau_f^A(c) = \frac{1-\rho_W}{4}\left[\left(\tau_A^{bc} - \tau_A^{bb}\right) + \left(\tau_A^{cc} - \tau_A^{cb}\right)\right]$$

change in the A-0 frame @ c from bulk when jumping @ bulk conditions

change in the A-0 frame @ c from bulk when jumping @ conc. conditions

$$\rightarrow \delta\tau_j^W(c) = \frac{1+\rho_W}{4}\left[\left(\tau_W^{cb} - \tau_W^{bb}\right) + \left(\tau_W^{cc} - \tau_W^{bc}\right)\right]$$

change in the jump from 0 @ c vs bulk when frame is @ bulk conditions.

change in the jump from 0 @ c vs bulk when frame is @ concentrated conditions.

$$\rightarrow \delta\tau_j^A(c) = \frac{1-\rho_W}{4}\left[\left(\tau_A^{cb} - \tau_A^{bb}\right) + \left(\tau_A^{cc} - \tau_A^{bc}\right)\right]$$

change in jump from A @ c vs bulk when frame is @ bulk conditions.

change in the jump from A @ c vs bulk when frame is @ conc. conditions.